

Schur product of additive cyclic codes

Burcu Ecem Karakaş¹, Ferruh Özbudak², and Buket Özkaya³

^{1,3}Institute of Applied Mathematics, Middle East Technical
University, Ankara, Türkiye*

²FENS, Faculty of Engineering and Natural Sciences, Sabancı
University, Istanbul, Türkiye†

April 29, 2026

Abstract

An important family of algebraic codes that is closely related to quasi-cyclic codes is additive cyclic codes. Using different representations, we investigate the Schur product of additive cyclic codes and explore their algebraic properties within this framework. In particular, we provide a study of an explicit polynomial generator matrix of the Schur product for additive cyclic codes of index 2 and 3, and then extend this study to arbitrary index, in terms of trace representation. Furthermore, we examine the distance properties of the Schur product of additive cyclic codes, extending BCH, HT and Hasse-Weil bounds known for cyclic codes.

Key Words : Additive cyclic codes, Schur product, BCH bound, HT bound, Hasse-Weil bound.

1 Preliminaries

Let q be a prime power and let $\mathbb{F}_q$ be the finite field with q elements. A linear code $\mathcal{C}$ is a subspace of $\mathbb{F}_q^m$, where $m \geq 1$. $\mathcal{C}$ has length m and dimension k , if $\dim_{\mathbb{F}_q}(\mathcal{C}) = k \leq m$. The minimum distance of $\mathcal{C}$ is denoted by $d = d(\mathcal{C})$, which is the minimum nonzero weight among its codewords. That is, $d(\mathcal{C}) = \min \{\text{wt}(c) : c \in \mathcal{C} \setminus \{0\}\}$ with $\text{wt}(c) = \#\{i \in \{0, \dots, m-1\} : c_i \neq 0\}$.

Let $m \geq 1$ be a positive integer such that $\gcd(m, q) = 1$ from this point on. Let r be the smallest integer satisfying $q^r \equiv 1 \pmod{m}$ and let $\mathbb{F}_{q^r}$ be the

*{burcuey, ozkayab}@metu.edu.tr

†ferruh.ozbudak@sabanciuniv.edu

extension of $\mathbb{F}_q$ such that $\mathbb{F}_{q^r} = \mathbb{F}_q(\xi)$, where $\xi \in \mathbb{F}_{q^r}$ is a primitive m^{th} root of unity (i.e., $\mathbb{F}_{q^r}$ is the splitting field of $x^m - 1$ over $\mathbb{F}_q$). Let $\mathcal{C}$ be an $[m, k]_q$ -linear code over $\mathbb{F}_q$ (i.e., $\mathcal{C}$ is a k -dimensional subspace of $\mathbb{F}_q^m$). The elements of $\mathcal{C}$ are the codewords $c = (c_0, c_1, \dots, c_{m-1})$. An $[m, k]_q$ -linear code $\mathcal{C}$ is a cyclic code if $c = (c_0, c_1, \dots, c_{m-1}) \in \mathcal{C}$ implies $(c_{m-1}, c_0, c_1, \dots, c_{m-2}) \in \mathcal{C}$. Let $R := \mathbb{F}_q[x]/\langle x^m - 1 \rangle$ and $g(x) \in R$ be the generator polynomial of the $[m, k]_q$ -cyclic code $\mathcal{C}$ with $\deg(g(x)) = m - k$. So, $\mathcal{C} = \langle g(x) \rangle$ is an ideal in R and $\mathcal{C}^\perp = \langle h^*(x) \rangle$, where $h(x) = \frac{x^m - 1}{g(x)}$ is a parity-check polynomial and its reciprocal is $h^*(x) = x^k h(\frac{1}{x})$. The q -cyclotomic coset of ξ^j is $[\xi^j] = \{\xi^j, \xi^{qj}, \dots, \xi^{q^{r-1}j}\}$ for any positive integer j . Let w, p, d, s be positive integers, we now set some notation:

- zero set of $\mathcal{C}$: $Z(\mathcal{C}) = \{\xi^j \in \mathbb{F}_{q^r} : g(\xi^j) = 0\} = \{\xi^{j_1}, \dots, \xi^{j_w}\}$,
- defining set of $\mathcal{C}$: $J = \{j_1, \dots, j_w\}$,
- generating set of $\mathcal{C}$: $I = \Omega_m \setminus J$, where $\Omega_m = \{0, 1, \dots, m-1\}$,
- zero set of $\mathcal{C}^\perp$: $Z(\mathcal{C}^\perp) = \{\xi^i \in \mathbb{F}_{q^r} : h^*(\xi^i) = 0\} = \{\xi^{i_1}, \dots, \xi^{i_p}\}$.

Theorem 1.1. *Let $\mathcal{C}$ be an $[m, k]_q$ -cyclic code over $\mathbb{F}_q$ and ξ be the primitive m^{th} root of unity, $\xi \in \mathbb{F}_{q^r}$ and $\gcd(m, q) = 1$. Assume that the defining set of $\mathcal{C}$ is $J = \{j_1, \dots, j_w\}$ and the generating set of $\mathcal{C}$ is $I = \Omega_m \setminus J = \{i_1, \dots, i_t\}$. Then the defining set of $\mathcal{C}^\perp$ is $-I = \{-i_1, \dots, -i_t\} = \{m - i : i \in I\}$.*

Now let us set the notion of defining set bounds. A nonempty subset $E \subseteq \Omega_m$ is said to be consecutive if there exist integers e, λ and δ with $e \geq 0, \delta \geq 2, \lambda \geq 1$ and $\gcd(m, \lambda) = 1$ such that $E = \{e + z\lambda : 0 \leq z \leq \delta - 2\} \subseteq \Omega_m$.

Let $\mathcal{P}(\Omega_m)$ be the power set of Ω_m . Note that any nonempty $P \in \mathcal{P}(\Omega_m)$ is the defining set of some cyclic code $\mathcal{D}_P \subseteq \mathbb{F}_{q^r}^m$ since $x^m - 1$ splits over $\mathbb{F}_{q^r}$. Let $\mathcal{C}$ be a cyclic code of length m over some subfield of $\mathbb{F}_{q^r}$ with nonempty defining set J , then for any nonempty $P \subseteq J$, $\mathcal{C}$ is a subfield subcode of $\mathcal{D}_P$ and we have $d(\mathcal{C}) \geq d(\mathcal{D}_P)$. We define a defining set bound to be member of the collection $\mathcal{B}(\mathcal{C}) := \{(P, d_P) : \emptyset \neq P \subseteq J\} \subseteq \mathcal{P}(\Omega_m) \times \mathbb{N}$ so that $P \subseteq J$ implies $d(\mathcal{C}) \geq d(\mathcal{D}_P) \geq d_P$, for any $(P, d_P) \in \mathcal{B}(\mathcal{C})$. Note that for any nonempty $P \subseteq J$, we have $(P, d_P) \in \mathcal{B}(\mathcal{C})$. We set the well-known BCH bound [BRC60], [Hoc59] as

$$\text{BCH}(\mathcal{C}) = \{(E, |E| + 1) \in \mathcal{B}(\mathcal{C}) : E \subseteq J \text{ is consecutive}\}.$$

Similarly, we formulate the HT bound [HT72] as

$$\text{HT}(\mathcal{C}) = \{(D, \delta + s) \in \mathcal{B}(\mathcal{C}) : D \subseteq J\},$$

where $D = \{e + z\lambda_1 + y\lambda_2 : 0 \leq z \leq \delta - 2, 0 \leq y \leq s\}$ for integers $e \geq 0, \delta \geq 2, s \geq 1, \lambda_1 \geq 1$ and $\lambda_2 \geq 1$ such that $\gcd(m, \lambda_1) = 1, \gcd(m, \lambda_2) < \delta$.

Using the defining set of the dual of a given cyclic code, we can represent any cyclic code as an evaluation code as follows.

Theorem 1.2. [Cas18, Lemma 4] Let $\mathcal{C}$ be an $[m, k]_q$ -cyclic code over $\mathbb{F}_q$, where $\gcd(m, q) = 1$. Then $B(-I)|_{\mathbb{F}_q} = B(-I) \cap \mathbb{F}_q^m = \mathcal{C}$ such that

$$\begin{aligned} B(-I) &= \left\{ (f(1), f(\xi), \dots, f(\xi^{m-1})) : f(x) \in \mathbb{F}_{q^r}[x], f(x) = \sum_{j \in -I} f_j x^j \right\} \\ &= \left\{ \left(\sum_{j \in -I} f_j \xi^{ij} \right)_{0 \leq i \leq m-1} : f_j \in \mathbb{F}_{q^r}, \forall j \right\}. \end{aligned}$$

Given two vectors $u = (u_0, \dots, u_{m-1}), v = (v_0, \dots, v_{m-1}) \in \mathbb{F}_q^m$, we define their *Schur product*, denoted by $u \star v$, as $u \star v = (u_0 \cdot v_0, \dots, u_{m-1} \cdot v_{m-1})$. In the polynomial representation over R , let $u(x) = \sum_{i \in \Omega} u_i x^i$ and $v(x) = \sum_{i \in \Omega} v_i x^i$, then $u(x) \star v(x) = \sum_{i \in \Omega} u_i v_i x^i$. If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two linear codes of the same length over the same field, then their Schur product is $\mathcal{C}_1 \star \mathcal{C}_2 := \text{span}_{\mathbb{F}_q} \{c_1 \star c_2 : c_1 \in \mathcal{C}_1, c_2 \in \mathcal{C}_2\}$. If G_1 and G_2 are generator matrices of $\mathcal{C}_1$ and $\mathcal{C}_2$ with k_1 and k_2 rows, respectively, then $\mathcal{C}_1 \star \mathcal{C}_2 = \text{span}_{\mathbb{F}_q} \{r_i \star s_j : 1 \leq i \leq k_1, 1 \leq j \leq k_2\}$ so that r_i and s_j denote the i^{th} and j^{th} rows of G_1 and G_2 , respectively.

For cyclic code pairs, we have the following result.

Theorem 1.3. [Cas18, Theorem 1] Assume that $\gcd(m, q) = 1$ and that $\mathcal{C}_1$ and $\mathcal{C}_2$ are two cyclic codes of length m over $\mathbb{F}_q$ with generating sets I_1 and I_2 , respectively. Let

$$\begin{aligned} B(-I_1) &= \left\{ \left(\sum_{j_1 \in -I_1} f_{j_1} \xi^{ij_1} \right)_{0 \leq i \leq m-1} : f_{j_1} \in \mathbb{F}_{q^r}, \forall j_1 \right\}, \\ B(-I_2) &= \left\{ \left(\sum_{j_2 \in -I_2} g_{j_2} \xi^{ij_2} \right)_{0 \leq i \leq m-1} : g_{j_2} \in \mathbb{F}_{q^r}, \forall j_2 \right\}. \end{aligned}$$

Then, $B(-(I_1 + I_2))|_{\mathbb{F}_q} = \mathcal{C}_1 \star \mathcal{C}_2$, where $I_1 + I_2 = \{i_1 + i_2 : i_1 \in I_1, i_2 \in I_2\}$.

We now generalize [Cas18, Definition 10] as follows. Given any $P \in \mathcal{P}(\Omega_m)$, the amplitude of P will be denoted by $\text{amp}(P)$ and it is defined as

$$\text{amp}(P) := \min \{ |E| : P \subseteq E \text{ and } E \text{ is consecutive} \}.$$

In other words, $\text{amp}(P)$ is the size of the smallest consecutive set in Ω_m that contains P . Note that, if $\text{amp}(I) = |E|$ such that I is the generating set of the cyclic code $\mathcal{C}$, for some consecutive set $E \subseteq \Omega_m$, then $\Omega_m \setminus E$ is the largest consecutive set contained in its defining set J . Hence, we have $(E, m - \text{amp}(I) + 1) \in \text{BCH}(\mathcal{C})$. We reformulate [Cas18, Theorem 2] as,

Theorem 1.4. *Let $\mathcal{C} = B(-I)|_{\mathbb{F}_q}$ be a cyclic code with generating set I . Then*

1. $\dim_{\mathbb{F}_q}(\mathcal{C}) = |I|$ and $\dim_{\mathbb{F}_q}(\mathcal{C} \star \mathcal{C}) = |I + I|$.
2. $d(\mathcal{C}) \geq m - \text{amp}(I) + 1$ and $d(\mathcal{C} \star \mathcal{C}) \geq m - \text{amp}(I + I) + 1$.

The above theorem is a direct result of Theorem 1.3 and the BCH bound.

Now we adopt a different notation than above, so that we are consistent with the works [Bie02], [GÖÖ17], [SLÖS22] and [SCÖ25].

Let $\mathbb{F}_{q^r}$ be the extension of the finite field $\mathbb{F}_q$, where r is a positive integer. Let $n|q^r - 1$ and $A = \{i_1, \dots, i_s\} \subseteq \mathbb{Z}/n\mathbb{Z}$ be arbitrary. Let α be a generator of the subgroup of order n in $\mathbb{F}_{q^r}$, say $W = \{\alpha^0, \dots, \alpha^{n-1}\}$. Define the space of polynomials $P(A)$ as

$$P(A) := \{a_1 x^{i_1} + \dots + a_s x^{i_s} : a_1, \dots, a_s \in \mathbb{F}_{q^r}\}$$

and

$$B(A) := \{(f(\alpha^0), \dots, f(\alpha^{n-1})) : f(x) \in P(A)\} \subset \mathbb{F}_{q^r}^n.$$

Let $m \leq r$ be a positive integer and $\tau = \{\gamma_1, \dots, \gamma_m\}$ be a $\mathbb{F}_q$ -linearly independent subset of $\mathbb{F}_{q^r}$. Let ϕ_τ be a surjective $\mathbb{F}_q$ -linear map from $\mathbb{F}_{q^r}$ to $\mathbb{F}_q^m$ such that $\phi_\tau(x) := (\text{Tr}(\gamma_1 x), \dots, \text{Tr}(\gamma_m x))$ where Tr is the trace map from $\mathbb{F}_{q^r}$ to $\mathbb{F}_q$. Then,

$$\begin{aligned} \phi_\tau(B(A)) &= \{\phi_\tau(f(\alpha^0), \dots, f(\alpha^{n-1})) : f(x) \in P(A)\} \subset \mathbb{F}_q^{mn} \\ &= \{(\text{Tr}(\gamma_1 f(\alpha^0)), \dots, \text{Tr}(\gamma_m f(\alpha^0)), \dots, \\ &\quad \text{Tr}(\gamma_1 f(\alpha^{n-1})), \dots, \text{Tr}(\gamma_m f(\alpha^{n-1}))) : f(x) \in P(A)\}, \end{aligned} \quad (1.1)$$

is an additive cyclic code of length n over $\mathbb{F}_q^m$ [GÖÖ17].

In [GÖÖ17], a lower bound on the minimum distance of certain additive cyclic codes was proven. To describe this bound, we let $\mathcal{C} = \phi_\tau(B(A))$, $n = q^r - 1$ and each $i_j \geq 1$ for $i_j \in A$. We define,

$$V := \{x \in \mathbb{F}_{q^r} : \text{Tr}(\gamma_1 x) = \dots = \text{Tr}(\gamma_m x) = 0\},$$

which is an $\mathbb{F}_q$ -linear subspace of $\mathbb{F}_{q^r}$ of dimension $r - m$. Then there exists a uniquely determined monic q -additive polynomial $F(y) = \sum_{i=0}^m F_i y^{q^i} \in \mathbb{F}_{q^r}[y]$ of degree q^m such that $V = \text{Im}(F)$ (see [GÖÖ17, Lemma 4]). Moreover, [GÖÖ17, Proposition 5] implies that $f(x) \in V$, for $x \in \mathbb{F}_{q^r}$, if and only if $F(y) = f(x)$, for some $y \in \mathbb{F}_{q^r}$. Hence, the weight of a codeword $c \in \mathcal{C}$ satisfies

$$\begin{aligned} \text{wt}(c) &= q^r - |\{x \in \mathbb{F}_{q^r} : \text{Tr}(\gamma_i, f(x)) = 0, \forall i \in \{1, \dots, m\}\}| \\ &= q^r - \frac{|\chi_f^{af}(\mathbb{F}_{q^r})|}{q^m}, \end{aligned} \quad (1.2)$$

where $\chi_f^{af}(\mathbb{F}_{q^r})$ denotes the set of affine $\mathbb{F}_{q^r}$ -rational points on the curve χ_f defined by $F(y) = f(x)$. The next result is derived from Serre's improvement on the Hasse-Weil bound [Sti09, Theorem 5.3.1] applied on (1.2).

Theorem 1.5. [GÖÖ17, Theorem 1] *Consider the additive cyclic code $\mathcal{C} = \phi_\tau(B(A))$ of length $n = q^r - 1$ over $\mathbb{F}_q^m$, where $A = \{i_1, \dots, i_s\} \subseteq \mathbb{Z}/n\mathbb{Z}$. Assume that $\gcd(i_j, q) = 1$ for all j and let $i = \max\{i_j : 1 \leq j \leq s\}$. Then,*

$$d(\mathcal{C}) \geq q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m}.$$

When $n|(q^r - 1)$, we obtain:

$$d(\mathcal{C}) \geq \frac{n}{q^r - 1} \left(q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m} \right),$$

where $i = \max \left\{ \frac{q^r - 1}{n} i_j \pmod{q^r - 1} : 1 \leq j \leq s \right\}$.

$\phi_\tau(B(A))$ given in (1.1) is an index m quasi-cyclic code. Therefore, they can be viewed as R -submodules of R^m , where $R = \mathbb{F}_q[x]/\langle x^n - 1 \rangle$.

2 Schur Product in Index 2

Assume that $\mathcal{C}$ is an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^2}$ of length n . Recall that such an additive cyclic code can be viewed as a quasi-cyclic code of index 2 and of length n and therefore the polynomial generator matrix of an additive cyclic code of length n over $\mathbb{F}_{q^2}$ is of the form (see [SLÖS22, Theorem 3.2], [SCÖ25, Theorem 3.1]),

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix}, \quad (2.1)$$

where $p(x), q(x), r(x) \in \mathbb{F}_q[x]/\langle x^n - 1 \rangle$ with $p(x)|(x^n - 1)$, $r(x)|(x^n - 1)$ and $\deg(q(x)) < \deg(r(x))$.

We want to find the generator matrix of $\mathcal{C} \star \mathcal{C}$, where $\star$ denotes the Schur product. Using the matrix (2.1), the polynomial generator matrix of $\mathcal{C} \star \mathcal{C}$ is:

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} \star \begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix},$$

which can be found as follows:

We define $f(x) \star f(x) = \sum_{i=0}^{n-1} f_i^2 x^i$, for any $f(x) \in \mathbb{F}_q[x]/\langle x^n - 1 \rangle$. Let

$$p^{(1)}(x) = \gcd(p(x) \star p(x), p(x) \star xp(x), \dots, p(x) \star x^{d_p} p(x)),$$

where $d_p = n - \deg(p(x)) - 1$ and let

$$r^{(1)}(x) = \gcd(r(x) \star r(x), r(x) \star xr(x), \dots, r(x) \star x^{d_r} r(x)),$$

where $d_r = n - \deg(r(x)) - 1$ by [Mir12, Proposition 4.2.2]. We have

$$p^{(1)}(x) = a_0(x)(p(x) \star p(x)) + \dots + a_{d_p}(x)(p(x) \star x^{d_p} p(x)),$$

for some $a_0(x), \dots, a_{d_p}(x) \in \mathbb{F}_q[x]/\langle x^n - 1 \rangle$, and

$$q^{(1)}(x) = a_0(x)(q(x) \star q(x)) + \dots + a_{d_p}(x)(q(x) \star x^{d_p} q(x)),$$

since $p(x)$ and $q(x)$ are in the same row. We obtain:

$$\begin{bmatrix} p^{(1)}(x) & q^{(1)}(x) \\ 0 & r^{(1)}(x) \end{bmatrix}.$$

Now, set $r^{(2)}(x)$ as follows:

If $d_p \geq d_r$, $r^{(2)}(x) = \gcd(r^{(1)}(x), q(x) \star r(x), xq(x) \star r(x), \dots, x^{d_p} q(x) \star r(x))$,

If $d_p < d_r$, $r^{(2)}(x) = \gcd(r^{(1)}(x), q(x) \star r(x), q(x) \star xr(x), \dots, q(x) \star x^{d_r} r(x))$.

It was observed in [Mir12] before Proposition 4.2.2 that $p^{(1)}(x)$ and $r^{(1)}(x)$ divide $(x^n - 1)$. Using this, we get

$$r^{(2)}(x) | r^{(1)}(x) \text{ and } r^{(1)}(x) | (x^n - 1) \Rightarrow r^{(2)}(x) | (x^n - 1).$$

Finally, we set

$$q^{(2)}(x) = q^{(1)}(x) - s_0(x)r^{(2)}(x),$$

such that $\deg(q^{(2)}(x)) < \deg(r^{(2)}(x))$. So,

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} \star \begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} = \begin{bmatrix} p^{(1)}(x) & q^{(2)}(x) \\ 0 & r^{(2)}(x) \end{bmatrix}.$$

Theorem 2.1. *Let $\mathcal{C}$ be an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^2}$ of length n with the polynomial generator matrix,*

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix}.$$

Then the Schur product $\mathcal{C} \star \mathcal{C}$ is also additive cyclic with the polynomial generator matrix

$$\begin{bmatrix} p^{(1)}(x) & q^{(2)}(x) \\ 0 & r^{(2)}(x) \end{bmatrix},$$

where $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x)$ can be found by using Algorithm 1.

Algorithm 1 Polynomial Generator Matrix Schur Product of Index 2

Input: $p(x), q(x), r(x)$

```

1:  $d_p = n - \deg(p(x)) - 1$ 
2:  $d_r = n - \deg(r(x)) - 1$ 
3:  $p^{(1)}(x) \leftarrow \gcd(p(x) \star p(x), p(x) \star xp(x), \dots, p(x) \star x^{d_p}p(x))$ 
4:  $r^{(1)}(x) \leftarrow \gcd(r(x) \star r(x), r(x) \star xr(x), \dots, r(x) \star x^{d_r}r(x))$ 
5:  $a_0(x), \dots, a_{d_p}(x) \leftarrow \mathbf{ExtendedEuclideanAlg}((p(x) \star p(x)), \dots, (p(x) \star x^{d_p}p(x)))$ 
6:  $q^{(1)}(x) \leftarrow a_0(x)(q(x) \star q(x)) + \dots + a_{d_p}(x)(q(x) \star x^{d_p}q(x))$ 
7: if  $d_p \geq d_r$  then
8:    $r^{(2)}(x) \leftarrow \gcd(r^{(1)}(x), q(x) \star r(x), xq(x) \star r(x), \dots, x^{d_p}q(x) \star r(x))$ 
9: else if  $d_p < d_r$  then
10:   $r^{(2)}(x) \leftarrow \gcd(r^{(1)}(x), q(x) \star r(x), q(x) \star xr(x), \dots, q(x) \star x^{d_r}r(x))$ 
11: end if
12:  $(s_0(x), q^{(2)}(x)) \leftarrow \mathbf{DivisionAlg}(q^{(1)}(x), r^{(2)}(x))$ 
13: return  $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x)$ 

```

Example 2.2. Let the polynomial generator matrix of $\mathcal{C}$ be

$$\begin{bmatrix} p(x) & q(x) \\ 0 & r(x) \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 1 \\ 0 & x - 1 \end{bmatrix}.$$

Therefore, the polynomial generator matrix of the Schur product $\mathcal{C} \star \mathcal{C}$ is

$$\begin{bmatrix} x^2 + 1 & 1 \\ 0 & x - 1 \end{bmatrix} \star \begin{bmatrix} x^2 + 1 & 1 \\ 0 & x - 1 \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Remark 2.3. If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two additive cyclic codes of the same length over $\mathbb{F}_{q^2}$, then their Schur product $\mathcal{C}_1 \star \mathcal{C}_2$ and the corresponding polynomial generator matrix can be computed in a similar way using Algorithm 1.

3 Schur Product in Index 3

Assume that $\mathcal{C}$ is a $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^3}$ of length n . Recall that such an additive cyclic code can be viewed as a quasi-cyclic code of index 3 and of length n . Therefore the polynomial generator matrix of an additive cyclic code of length n over $\mathbb{F}_{q^3}$ is of the form (see [SCÖ25, Theorem 3.2]),

$$\begin{bmatrix} p(x) & q(x) & m(x) \\ 0 & r(x) & n(x) \\ 0 & 0 & l(x) \end{bmatrix}, \quad (3.1)$$

where $p(x), q(x), r(x), m(x), n(x), l(x) \in \mathbb{F}_q[x]/\langle x^n - 1 \rangle$ with $p(x)|(x^n - 1)$, $r(x)|(x^n - 1)$, $l(x)|(x^n - 1)$, $\deg(q(x)) < \deg(r(x))$, $\deg(m(x)) < \deg(l(x))$ and $\deg(n(x)) < \deg(l(x))$.

Theorem 3.1. *Let $\mathcal{C}$ be an $\mathbb{F}_q$ -linear additive cyclic code over $\mathbb{F}_{q^3}$ of length n with the polynomial generator matrix,*

$$\begin{bmatrix} p(x) & q(x) & m(x) \\ 0 & r(x) & n(x) \\ 0 & 0 & l(x) \end{bmatrix}.$$

Then the Schur product $\mathcal{C} \star \mathcal{C}$ is also additive cyclic with the polynomial generator matrix

$$\begin{bmatrix} p^{(1)}(x) & q^{(2)}(x) & m^{(3)}(x) \\ 0 & r^{(2)}(x) & n^{(3)}(x) \\ 0 & 0 & l^{(3)}(x) \end{bmatrix},$$

where $p^{(1)}(x), q^{(2)}(x), r^{(2)}(x), m^{(3)}(x), n^{(3)}(x), l^{(3)}(x)$ can be found by using Algorithm 2 which will be provided in the full version.

Example 3.2. *Let the polynomial generator matrix of $\mathcal{C}$ be*

$$\begin{bmatrix} p(x) & q(x) & m(x) \\ 0 & r(x) & n(x) \\ 0 & 0 & l(x) \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 1 & 2x + 1 \\ 0 & x - 1 & x \\ 0 & 0 & x^2 + 1 \end{bmatrix}.$$

Hence, the polynomial generator matrix of the Schur product $\mathcal{C} \star \mathcal{C}$ is

$$\begin{bmatrix} x^2 + 1 & 1 & 2x + 1 \\ 0 & x - 1 & x \\ 0 & 0 & x^2 + 1 \end{bmatrix} \star \begin{bmatrix} x^2 + 1 & 1 & 2x + 1 \\ 0 & x - 1 & x \\ 0 & 0 & x^2 + 1 \end{bmatrix} = \begin{bmatrix} x^2 + 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Remark 3.3. *If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two additive cyclic codes of the same length over $\mathbb{F}_{q^3}$, then their Schur product $\mathcal{C}_1 \star \mathcal{C}_2$ and the corresponding polynomial generator matrix can be computed in a similar way.*

In this polynomial representation, it would become too complicated to express the Schur product for index 4 or higher indices. Therefore, we use the trace representation of the additive cyclic codes introduced by Bierbrauer [Bie02], which is valid for an arbitrary index.

4 Schur Product in Arbitrary Index

Recall that additive cyclic codes can be represented as (1.1). Note that if $m = 1$ and A is Galois closed, then by taking $A = -I$ and using Delsarte's theorem [Bie02, Theorem 4], we can apply Theorem 1.3 for the Schur product of cyclic codes. Let $m \geq 1$ and A be arbitrary. We want to characterize $\phi_\tau(B(A)) \star \phi_\tau(B(A))$. Take any two codewords from $\phi_\tau(B(A))$, which have coordinates $\text{Tr}(\gamma_i f(\alpha^j))$ and $\text{Tr}(\gamma_i g(\alpha^j))$, where $f(x), g(x) \in P(A)$ such that $1 \leq i \leq m$ and $1 \leq j \leq n - 1$. We have

$$\text{Tr}(\gamma_i f(\alpha^j)) = \gamma_i f(\alpha^j) + \gamma_i^q (f(\alpha^j))^q + \cdots + \gamma_i^{q^{r-1}} (f(\alpha^j))^{q^{r-1}},$$

and

$$\text{Tr}(\gamma_i g(\alpha^j)) = \gamma_i g(\alpha^j) + \gamma_i^q (g(\alpha^j))^q + \cdots + \gamma_i^{q^{r-1}} (g(\alpha^j))^{q^{r-1}}.$$

Then we have found that

$$\phi_\tau(B(A)) \star \phi_\tau(B(A)) \subseteq \phi_\tau(B(A + \bar{A})),$$

where $A + \bar{A} = \{i_{j_1} + q^\ell i_{j_2} : 0 \leq \ell \leq r', 1 \leq j_1, j_2 \leq s\} \subseteq \bar{A} + \bar{A} \subseteq \tilde{A} + \tilde{A}$, such that $\bar{A} := \{q^\ell i_j : 0 \leq \ell \leq r', 1 \leq j \leq s\}$ and the Galois closure of A is $\tilde{A} := \{q^\ell i_j : 0 \leq \ell \leq r-1, 1 \leq j \leq s\}$.

As a result, the Schur product of additive cyclic codes are again additive cyclic and

$$\phi_\tau(B(A)) \star \phi_\tau(B(A)) \subseteq \phi_\tau(B(A + \bar{A})) \subseteq \phi_\tau(B(\tilde{A} + \tilde{A})).$$

Note that equality holds when $A = \tilde{A}$, i.e., A is Galois closed.

Theorem 4.1. *Let $\phi_\tau(B(A))$ be an additive cyclic code of length n over $\mathbb{F}_q^m$, then the Schur product $\phi_\tau(B(A)) \star \phi_\tau(B(A))$ is again additive cyclic and it is defined as*

$$\phi_\tau(B(A)) \star \phi_\tau(B(A)) = \{\phi_\tau(h(\alpha^0), \dots, h(\alpha^{n-1})) : h(x) \in P(A + \bar{A})\}$$

such that $\phi_\tau(B(A)) \star \phi_\tau(B(A)) \subset \phi_\tau(B(A + \bar{A}))$ with

$$\begin{aligned} h(x) = & \gamma_i^1 f(x)g(x) + \gamma_i^q (f(x)g(x))^q + f(x)^q g(x) \\ & + \cdots + \gamma_i^{q^{r'}} (f(x)g(x))^{q^{r'}} + f(x)^{q^{r'}} g(x), \end{aligned}$$

where $f(x), g(x) \in P(A)$, and

$$r' = \begin{cases} \frac{r-1}{2}, & \text{if } r \text{ is odd,} \\ \frac{r}{2}, & \text{if } r \text{ is even.} \end{cases}$$

Remark 4.2. *We have $|\bar{A}| \leq r'|A|$ and $|\tilde{A}| \leq r|A|$, and therefore $|\bar{A} + \bar{A}| \leq r'|A + A|$. If $\mathcal{C}_1$ and $\mathcal{C}_2$ are two additive cyclic codes of the same length over $\mathbb{F}_q^m$, then their Schur product $\mathcal{C}_1 \star \mathcal{C}_2$ and the corresponding polynomial $h(x)$ can be computed in a similar way.*

5 Distance and Dimension of Schur Product

We assume the notation in Section 1. Our aim is to find estimates on $d(\mathcal{C} \star \mathcal{C})$ and $d((\mathcal{C} \star \mathcal{C})^\perp)$, for given $A \subseteq \Omega_n = \mathbb{Z}/n\mathbb{Z}$ and $\mathcal{C} = \phi_\tau(B(A))$.

For a given additive cyclic code $\mathcal{C} = \phi_\tau(B(A))$, it was shown in [Bie02, Theorem 23] that $d(\mathcal{C}^\perp) \geq |E| + 1$, where E is a consecutive set contained in A (i.e., $(E, |E| + 1) \in \text{BCH}(\mathcal{C}^\perp)$). Note that $\Omega_n \setminus E = \text{amp}(\Omega_n \setminus A)$, if E is the largest consecutive set contained in A . We get the following (cf. Theorem 1.4).

Theorem 5.1. *Let $\mathcal{C} = \phi_\tau(B(A))$ be an $\mathbb{F}_q$ -linear additive cyclic code of length n over $\mathbb{F}_q^m$. Let E and $\bar{E}$ be the largest consecutive sets contained in A and $\bar{A}$, respectively. Then*

1. $\dim_{\mathbb{F}_q}(\mathcal{C}) \geq |E|$ and $\dim_{\mathbb{F}_q}(\mathcal{C} \star \mathcal{C}) \geq |E| + |\bar{E}|$.
2. $d(\mathcal{C}^\perp) \geq n - \text{amp}(\Omega_n \setminus A) + 1$ and $d((\mathcal{C} \star \mathcal{C})^\perp) \geq n - \text{amp}(\Omega_n \setminus (A + \bar{A})) + 1$.
In particular, $(E, |E| + 1) \in \text{BCH}(\mathcal{C}^\perp)$ and $(E + \bar{E}, |E| + |\bar{E}| + 1) \in \text{HT}((\mathcal{C} \star \mathcal{C})^\perp)$.

Proof. Part (1) follows from the Singleton bound, which asserts $d \leq n - k - 1$ [LX04, Theorem 5.4.1], where $d = d(\mathcal{C})$ and $k = \dim_{\mathbb{F}_q}(\mathcal{C})$. Applying on $\mathcal{C}^\perp$, we get $d(\mathcal{C}^\perp) \leq n - (n - k) + 1 = k + 1$, which implies $k \geq d(\mathcal{C}^\perp) - 1$. Using [GÖÖ17, Theorem 2], we obtain $\dim_{\mathbb{F}_q}(\mathcal{C}) \geq d(\mathcal{C}^\perp) - 1 \geq |E|$. If we define $\bar{E} \subseteq \bar{A}$ as the largest consecutive set contained in $\bar{A}$, we get $\dim_{\mathbb{F}_q}(\mathcal{C} \star \mathcal{C}) \geq d((\mathcal{C} \star \mathcal{C})^\perp) - 1 \geq |E| + |\bar{E}|$.

Part (2) follows from the observation that $\Omega_n \setminus E = \text{amp}(\Omega_n \setminus A)$ and $|\Omega_n \setminus E| = n - |E|$. Then we apply the same idea on $B(A + \bar{A})^\perp$, which is characterized by $A + \bar{A}$. $\square$

Now we generalize Theorem 1.5 in Section 1 to the Schur product $\mathcal{C} \star \mathcal{C}$ of the given additive cyclic code $\mathcal{C} = \phi_\tau(B(A))$.

Theorem 5.2. *Let $n = q^r - 1$ and $A = \{i_1, \dots, i_s\} \subseteq \Omega_n$. Assume that $\gcd(i_j, q) = 1$ for all j , then $\gcd(i_{j_1} + q^\ell i_{j_2}, q) = 1$, $\forall j_1, j_2 \in \{1, \dots, s\}$ and $0 \leq \ell \leq r'$. Let i be the maximum of $A + \bar{A} = \{i_{j_1} + q^\ell i_{j_2} : 0 \leq \ell \leq r', 1 \leq j_1, j_2 \leq s\}$, then*

$$d(\mathcal{C} \star \mathcal{C}) \geq q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m}.$$

If $n|(q^r - 1)$, then we obtain

$$d(\mathcal{C} \star \mathcal{C}) \geq \frac{n}{q^r - 1} \left(q^r - q^{r-m} - \frac{(q^m - 1)(i - 1)\lfloor 2\sqrt{q^r} \rfloor}{2q^m} \right),$$

where $i = \max \left\{ \frac{q^r - 1}{n} (i_{j_1} + q^\ell i_{j_2}) \pmod{q^r - 1} : 0 \leq \ell \leq r', 1 \leq j_1, j_2 \leq s \right\}$.

References

- [Bie02] Jürgen Bierbrauer. The theory of cyclic codes and a generalization to additive codes. *Designs, Codes and Cryptography*, 25:189–206, 2002.
- [BRC60] Raj Chandra Bose and Dwijendra K Ray-Chaudhuri. On a class of error correcting binary group codes. *Information and control*, 3(1):68–79, 1960.

- [Cas18] Ignacio Cascudo. On squares of cyclic codes. *IEEE Transactions on Information Theory*, 65(2):1034–1047, 2018.
- [GÖÖ17] Cem Güneri, Ferruh Özbudak, and Funda Özdemir. Hasse–Weil bound for additive cyclic codes. *Designs, Codes and Cryptography*, 82(1):249–263, 2017.
- [Hoc59] Alexis Hocquenghem. Codes correcteurs d’erreurs. *Chiffers*, 2:147–156, 1959.
- [HT72] Carlos RP Hartmann and Kenneth K Tzeng. Generalizations of the BCH bound. *Information and control*, 20(5):489–498, 1972.
- [LX04] San Ling and Chaoping Xing. *Coding theory: a first course*. Cambridge university press, 2004.
- [Mir12] Diego Mirandola. Schur products of linear codes: a study of parameters. Master’s thesis, Dept. Math. Sci., Stellenbosch Univ., Stellenbosch, South Africa, 2012.
- [SCÖ25] Minjia Shi, Shixiao Chu, and Ferruh Özbudak. Additive cyclic codes over $\mathbb{F}_{q^3}$. *Journal of Algebra and its Applications*, 24(09):2550339, 2025.
- [SLÖS22] Minjia Shi, Na Liu, Ferruh Özbudak, and Patrick Solé. Additive cyclic complementary dual codes over $\mathbb{F}_4$. *Finite Fields and Their Applications*, 83:102087, 2022.
- [Sti09] Henning Stichtenoth. *Algebraic function fields and codes*. Springer, 2009.